PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

DAR ES SALAAM REGION

FORM FOUR - MOCK EXAMINATION - 2025

PHYSICS 2A (ACTUAL PRACTICAL)

CODE:031/2A
TIME:2:30Hours

Wednesday 23rd July 2025 A.M

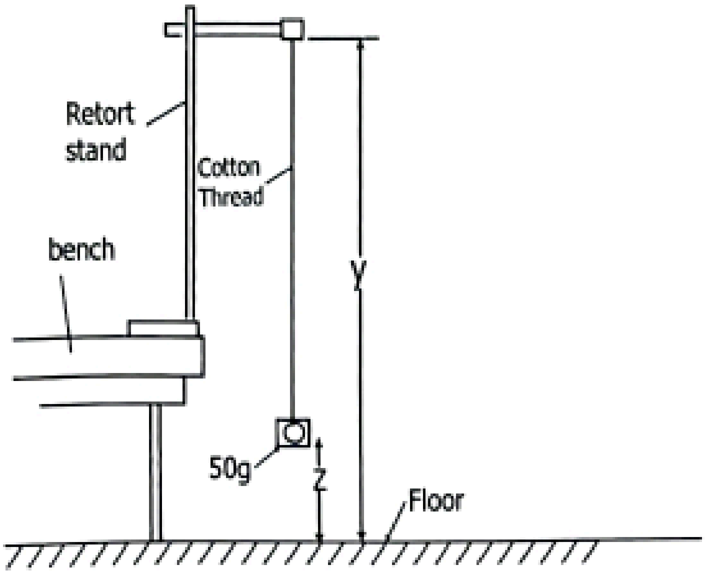
INSTRUCTIONS

1. This paper consists of two **(2)** questions answers **all** two questions
2. Each question carries a total of twenty five **(25) marks**
3. All writings should be in **blue** or **black** ink, except for diagrams which must be in pencil
4. Mathematical tables and non-program able calculator may be used
5. Communication device are **not** allowed in the examination room.

This paper consists of 3 printed pages

1. You are provided with the following apparati, report stand 50c standard mass, meter rule, stop watch and cotton thread

Procedure



(a) Tie the standard mass of 50g on the cotton thread

(b) Hang the standard mass with cotton thread on the report stand so that the length 'Y' is equal to 110cm.from the floor up to the point of suspension and distance Z from the floor to the standard mass is 18cm.

(c) Displace the standard mass at a small angle and release it. Record time 't' for 20 isolations

(d) Respect the above procedure when varying Z by 18cm to obtain more four readings

(i) Tabulate your results for values of Z, (Y-Z), t(s), T^2 (s) and T^2 (s^2)

(ii) Plot a graph of (Y-Z)cm against T^2 (s^2)

(iii) Find gradient from the graph

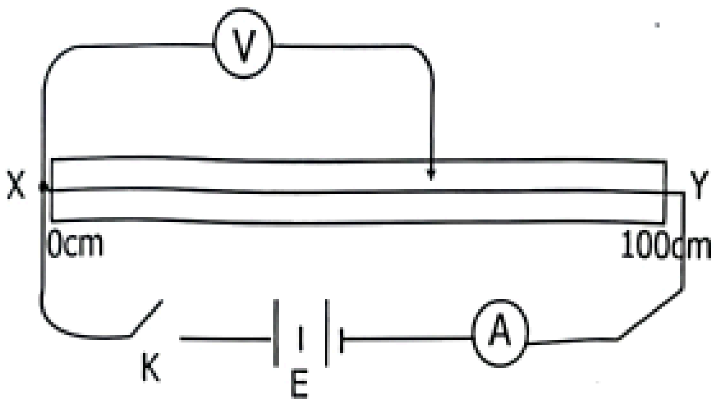
(iv) Determine the acceleration due to gravity given that

$$Z = \frac{4\pi^2 Y - gT^2}{4\pi^2}$$

(v) State any three possible sources of errors and its precaution to be **(25 marks)**

2. You are provided with potentiometer, a dry cells ammeter, voltmeter, jockey, key and connecting wires.

Procedure.



Procedure.

(i) Connect up the circuit as shown above

(ii) Close the key, put a jockey at length 'I' equal to 10cm.Read and record the voltage, V and correct I

(iii) Repeat the above procedure (II) for charge the values of length 'L' equal to 20cm, 30cm, 40cm and 50cm, each time read and record values of voltages and current

(a) Tabulate your results for values of L(m) V (v) and I (A)

(b) i) Plot a graph of V against l (cm)

ii) Determine slope S of the graph

(c) The graph use the equation V=KL

i) What the value of K ?

ii) What does K means ?

(d) Determine resistance, R of the wire of potentiometer given at a length of 80cm.

(e) Suggest the air of this experiment **(25 marks)**

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PHYSICS 2A

MARKING SCHEME

Q1. (i) Table of results.

Z (cm)	(Y-Z) (cm)	t (s)	T (s)	T^2 (s^2)
18	92	36.1	1.81	3.28
36	74	32.23	1.61	2.59
54	56	27.45	1.37	1.88
72	38	22.87	1.14	1.3
90	20	16.02	0.40	0.64

10% 00½ @

iii)

$$\text{slope} = \frac{\Delta (Y-Z)}{\Delta T^2}$$ 00½

$$\text{slope} = \left(\frac{68-50}{2.4-1.68}\right) cm/s^2$$ 00½

$$\text{slope} = 25\ cm/s^2$$ 01

1...

iv) Acceleration due to gravity, "g".

From $z = \frac{4\pi^2 y - gT^2}{4\pi^2}$

$$4\pi^2 z = 4\pi^2 y - gT^2$$

$$gT^2 = 4\pi^2 y - 4\pi^2 z$$

$$gT^2 = 4\pi^2 (y - z)$$

$$\frac{gT^2}{4\pi^2} = y - z.$$

$$y - z = \frac{gT^2}{4\pi^2}$$

$$y = mx + c$$

$$\text{slope} = g/4\pi^2$$

$$g = 4\pi^2 \times \text{slope}$$

$$g = 4\pi^2 \times 25\,cm/s^2$$

$$g = 986.96\,cm/s^2$$

$$\underline{g = 9.8696\,m/s^2}$$

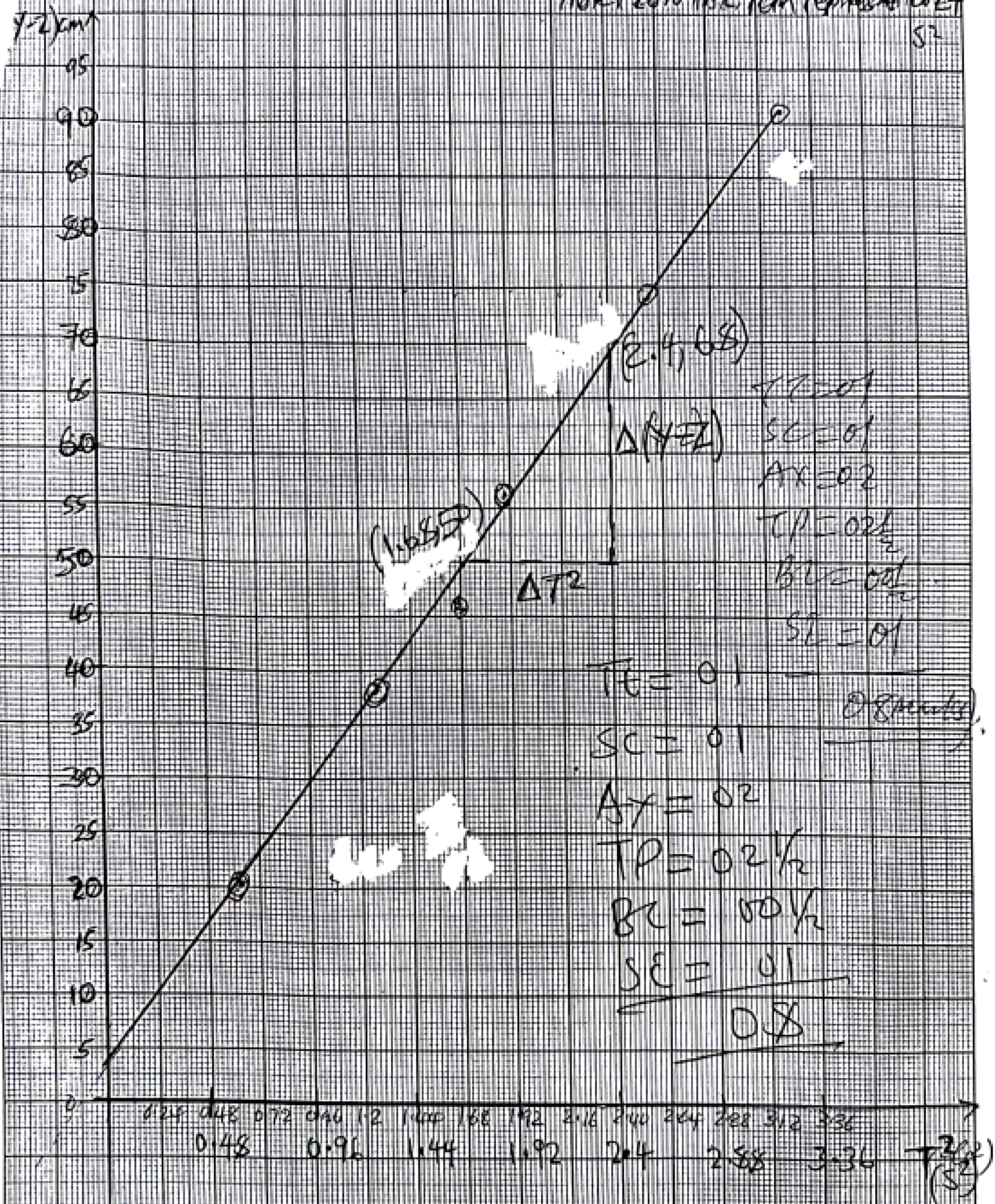
1 QN 1 (ii)

A GRAPH OF (Y-Z)cm AGAINST T^2 (s^2)

SCALE.

VERTICAL: 1cm represents 5cm

HORIZONTAL: 1cm represents 0.24 s^2



(1cm x 1cm) squares of (10mm x 10mm)

Qn 2.

(a) Table of result 00½ a. 07½

Length L (m)	Potential differ V(V)	Current, I A
0.1	0.22	0.28
0.2	0.4	0.28
0.3	0.6	0.28
0.4	0.8	0.28
0.5	1.0	0.28

b) ii) Slope, $S = \frac{\Delta V}{\Delta L}$ 00½

$$\text{slope} = \frac{0.9 - 0.3}{0.45 - 0.15} = \frac{0.6}{0.3}$$ 01

$$\text{Slope} = 2 V/m.$$ 02

c) i) The value of K is equal to the slope of ~~the~~ graph, that is 2 V/m. 01

ii) K in the equation V = KL represent a potential fall or potential gradient per ~~ ~~ unit length of wire. 01

d) Resistance of wire at 80 cm

$R = \frac{V}{I}$ $\quad I = 0.28A.$ 00½

V = voltage across a wire at 80 cm.

Since $2V / 100\,cm$, then for 80 cm

$\frac{2V \times 80\,cm}{100\,cm} = \underline{1.6V}$ 01

$R = \frac{1.6V}{0.28A}$

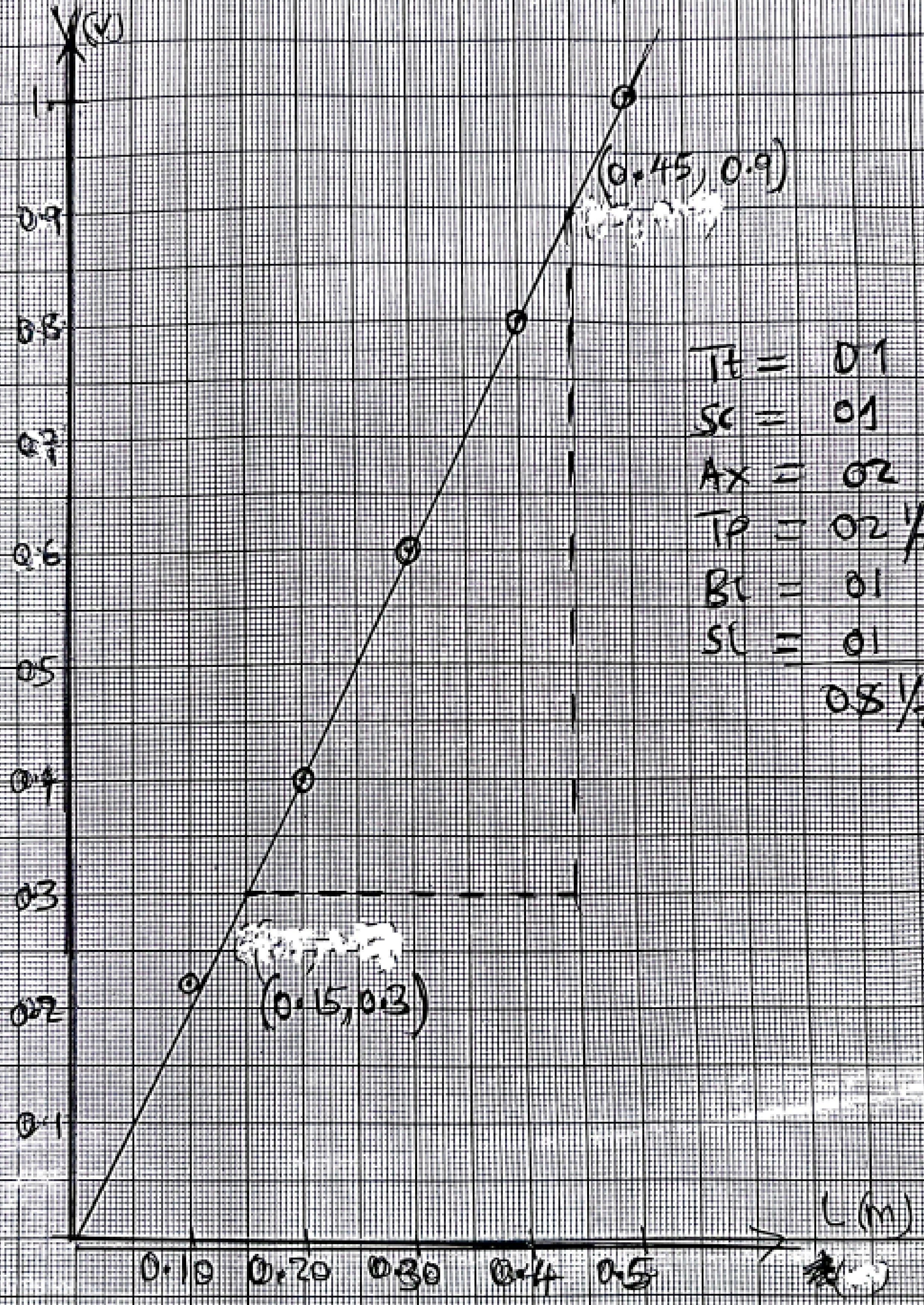
$R = 5.71\Omega.$ 02

Qn 2(i)

A GRAPH OF V AGAINST L

VS = 1cm represent 0.05V

HS = 1cm represent 5cm



Tt = 01

Sc = 01

Ax = 02

TP = 02 ½

Bl = 01

Sl = 01

08 ½

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(1cm x 1cm) squares of (10mm x 10mm)